

# AMAR DHILLON

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## Education

### University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028

College Park, MD

## Experience

### Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java policies that tune upper/lower bounds on EC2 capacity reserved for Amazon Redshift operations like patching, balancing availability against cost; initial backtests complete

### Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Rebuilt a legacy test-data workflow builder in React/TypeScript + Electron, adding a React-Flow editor and Jest-tested validation; **90%** of 30+ surveyed developers preferred it to the old tool
- Precomputed a category-keyed tree index of popular workflows from a 270K+ row PostgreSQL table (Python/SQLAlchemy, Redis cache) to minimize DB queries under access limits
- Cut LLM workflow generation time **65%** to under 15s by decomposing a Claude skill into sub-skills backed by Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

### UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Shipped a space-research tool used by Amazon Leo's public-policy team, now maintained by Leo engineers; deployed on AWS Fargate (Docker) via GitHub Actions and CDK (TypeScript)
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated a production EventBridge fine-tuning loop for a Bedrock document-importance classifier, retraining only when a larger LLM's cross-model eval scores move past a threshold

### UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built a Python pipeline for PhysTwin (PhD-led digital twin, targeting ICLR) turning AnyGrasp output on multi-camera images into Open3D overlays for a VLM; tuned its prompts to pick correct grasps and judge depth
- Shortened research-environment setup time **30%** with uv dependency pinning and Docker; added psutil time/memory profiling and Pytest suites to find bottlenecks

### TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member campus product studio that has shipped 5 live products, including Tortuga, a class scheduler with **20,000+ visits** in its first 30 days; built Redis caching for its Prisma ORM queries
- Building a RAG meal-suggestion feature for the official UMD app (GPT-4o mini, ChromaDB + SentenceBERT search); **93%** of generated meals rated satisfactory by internal testers

## Projects

### IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Engineered a C++17 order book and matching engine that replayed **50.9M** real IEX DEEP+ messages with no resync, matching 99,122 of 99,123 fills exactly and 100% of top-of-book prices
- Benchmarked B+ tree vs LSM skip-list stores for the simulation's event logs on identical 2M-record workloads; the B+ tree gave **12.8×** read throughput and **22×** faster range summaries for 27% more disk
- Sustained **1.26M events/sec** end-to-end over the full replay, averaging 686.7 ns/event in the matching engine

## Technical Skills

**Languages:** Python, Java, C++, C, TypeScript/JavaScript, SQL

**Cloud (AWS):** Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS

**Web & Data:** React, Electron, Node.js, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

**AI/ML & Tools:** PyTorch, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf

**Certification:** AWS Certified Developer – Associate (Nov 2024 – Nov 2027)